

Gender	Car Ownership	Travel cost	Income Level	Transportation
M	0	CHEAP	LOW	BUS
M	1	CHEAP	MEDIUM	BUS
F	1	CHEAP	MEDIUM	TRAIN
F	0	CHEAP	LOW	BUS
M	1	CHEAP	MEDIUM	BUS
M	0	STANDARD	MEDIUM	TRAIN
F	1	STANDARD	MEDIUM	TRAIN
F	1	EXPENSIVE	HIGH	CAR
M	2	EXPENSIVE	MEDIUM	CAR
F	2	EXPENSIVE	HIGH	CAR

Classify the following samples

Gender	Car Ownership	Travel cost	Income Level	Transportation
M	1	STANDARD	HIGH	?
M	0	CHEAP	MEDIUM	?
F	1	CHEAP	HIGH	?

Class probability

Bus --- $4/10 = 0.4$

- Train --- $3/10 = 0.3$

- Car --- $3/10 = 0.3$

- Info (D) or Entropy = $-0.4 \log_2(0.4) - 0.3 \log_2(0.3) - 0.3 \log_2(0.3) = 1.571$

a. Attribute Gender

Gender	Transportation
M	B
M	B
M	B
M	T
M	C

Gender	Transportation
F	T
F	B
F	T
F	C
F	C

Gender	Bus	Train	Car	$I(B_i, T_i, C_i)$
Male	3/5	1/5	1/5	1.371
Female	1/5	2/5	2/5	1.522

$$\text{Info}_{\text{gender}}(D) = 5/10 * 1.371 + 5/10 * 1.522 = 1.451$$

$$\begin{aligned} \text{Gain}(\text{gender}) &= \text{Info}(D) - \text{Info}_{\text{gender}}(D) \\ &= 1.571 - 1.451 = 0.12 \end{aligned}$$

b. Attribute Car ownership

Car ownership	Bus	Train	Car	I(Bi,Ti,Ci)
0	2/3	1/3	0/3	0.918
1	2/5	2/5	1/5	1.522
2	0	0	2/2	0

$$\text{Info}_{\text{car_owner}}(D) = 3/10 * 0.918 + 5/10 * 1.522 + 2/10 * 0 = 1.037$$

$$\text{Gain}(\text{car_owner}) = \text{Info}(D) - \text{Info}_{\text{car_owner}}(D) \\ 1.571 - 1.037 = 0.534$$

C. Travel cost

Travel Cost	Bus	Train	Car	I(Bi,Ti,Ci)
Cheap	4/5	1/5	0/5	0.722
Standard	0	2/2	0	0
Expensive	0	0	3/3	0

$$\text{Info}_{\text{Travel_cost}}(D) = 5/10 * 0.722 + 2/10 * 0 + 3/10 * 0 = 0.361$$

$$\text{Gain}_{\text{Travel_cost}} = \text{Info}(D) - \text{Info}_{\text{Travel_cost}}(D) \\ 1.571 - 0.361 = 1.21$$

d. Income Level

Income Level	Bus	Train	Car	I(Bi,Ti,Ci)
Low	2/2	0/2	0/2	0
Medium	2/6	3/6	1/6	1.459
High	0	0	2/2	0

$$\text{Info}_{\text{Income_level}}(D) = 2/10 * 0 + 6/10 * 1.459 + 2/10 * 0 = 0.876$$

$$\text{Gain}(\text{Income_level}) = \text{Info}(D) - \text{Info}_{\text{Income_level}}(D) \\ 1.571 - 0.876 = 0.695$$

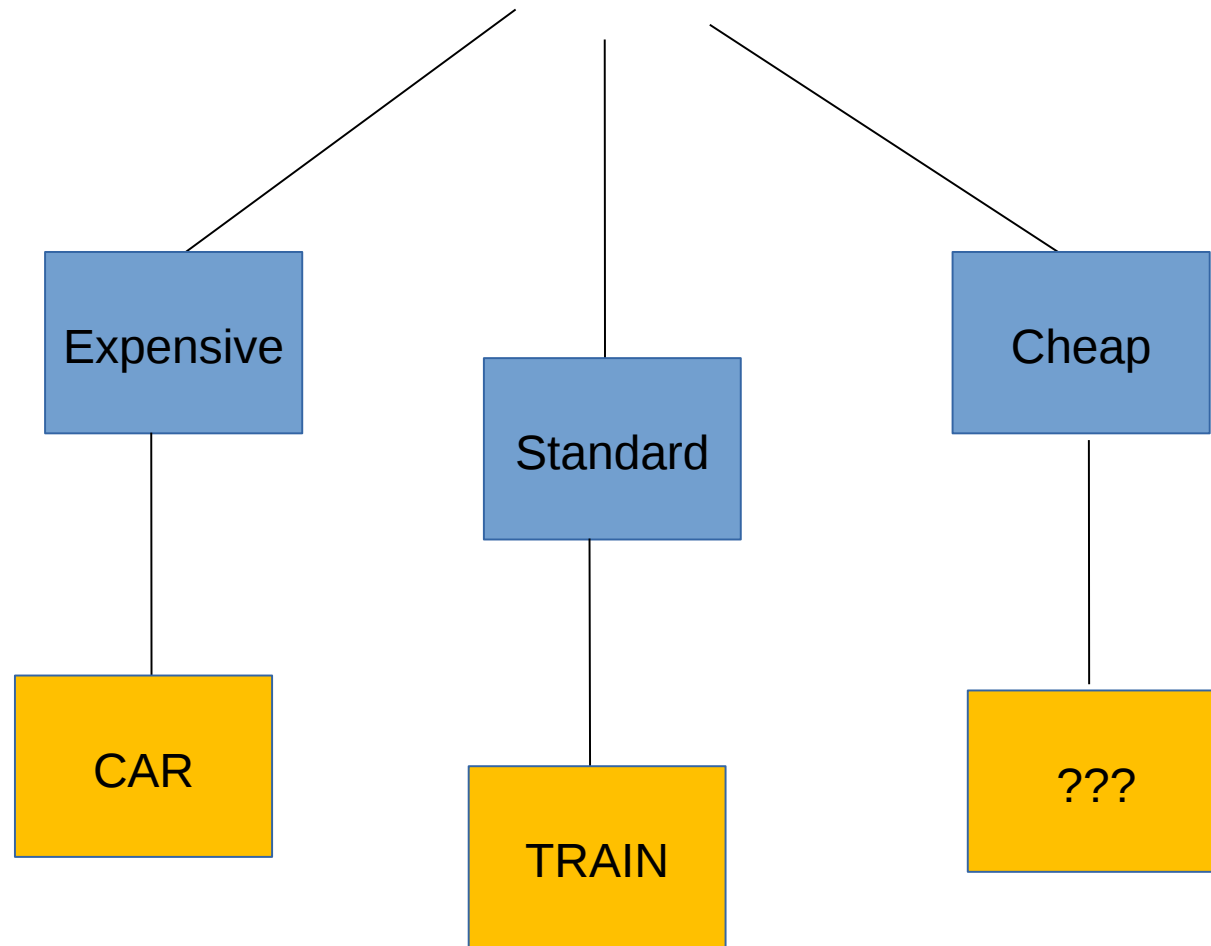
Choose High Gain Attribute

Attribute	Gain
Gender	0.12
Car	0.534
Travel cost	1.21
Income level	0.695

Travel cost

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Travel Cost



II Iteration

Attribute Gender, Car, Income Level

Gender	Car	Income Level	Transport
M	0	L	B
M	1	M	B
F	1	M	T
F	0	L	B
M	1	M	B

Probability Bus --- 4/5
Train --- 1/5

$$\text{Info}(D) = -4/5 \log_2 4/5 - 1/5 \log_2 1/5$$

$$\text{Info}(D) = 0.722$$

Gender

Gender	Transport		
M	B		
M	B		
M	B		
F	T		
F	B		
M	3/3	0	0
F	1/2	1/2	1

$$\text{Info}_{\text{gender}}(D) = 3/5 * 0 + 2/5 * 1 = 0.4$$

$$\text{Gain}_{\text{gender}}(D) = 0.722 - 0.4 = 0.322$$

Car ownership

Car ownership	Transport		
0	B		
0	B		
1	B		
1	T		
1	B		
car_ow ner	B	T	I(Bi,Ti)
0	2/2	0	0
1	2/3	1/3	0.918

$$\text{Info}_{\text{car_owner}}(D) = 0 + \frac{3}{5} * 0.918 = 0.5508$$

$$\text{Gain}_{\text{gender}}(D) = 0.722 - 0.5508 = 0.1712$$

Income level

Income level	Transport		
L	B		
L	B		
M	B		
M	T		
M	B		
Income_level	B	T	I(Bi,Ti)
Low	2/2	0	0
Med	2/3	1/3	0.918

$$\text{Info}_{\text{Income_level}}(D) = 0 + 3/5 * 0.918 = 0.5508$$

$$\text{Gain}_{\text{Income_level}}(D) = 0.722 - 0.5508 = 0.1712$$

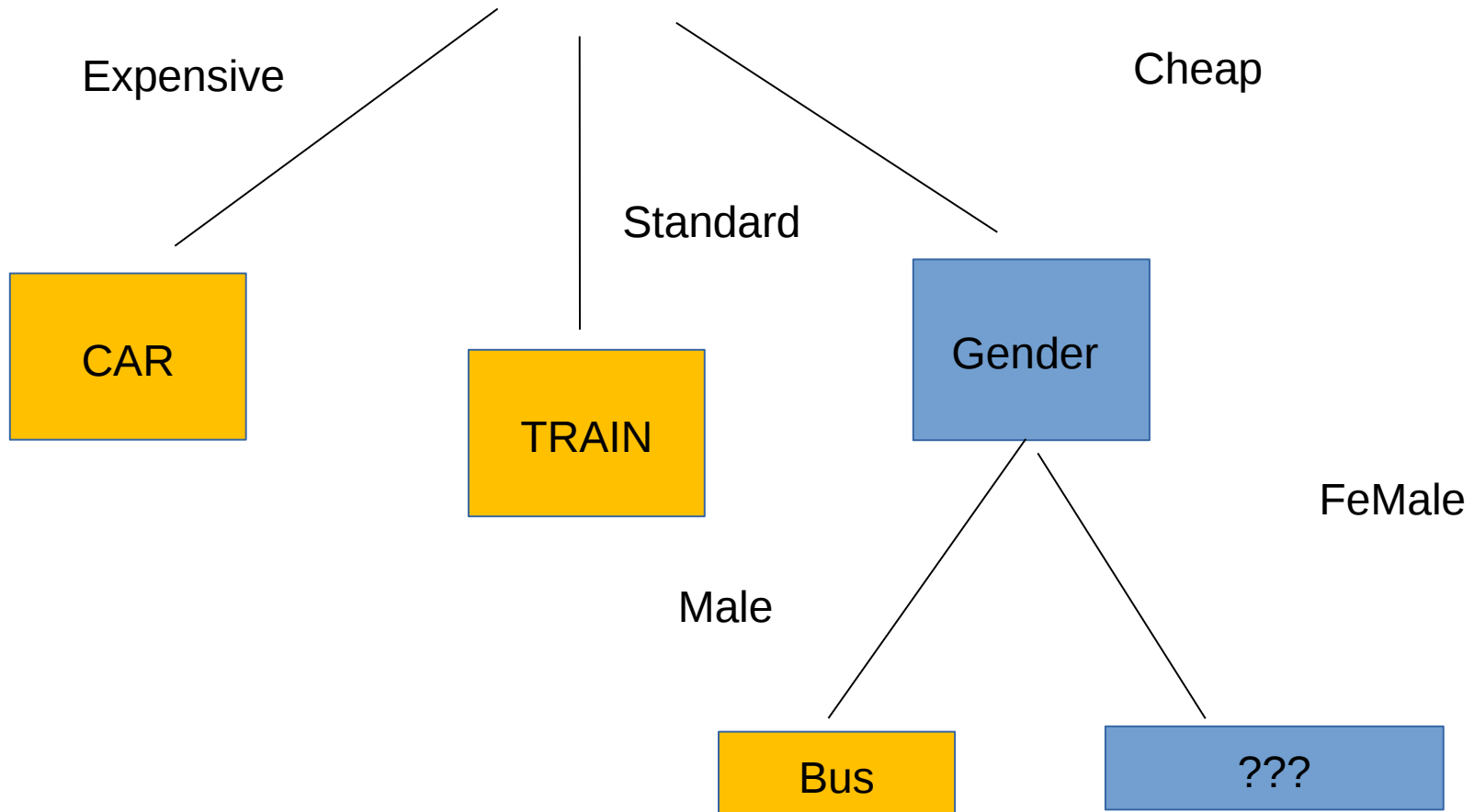
Choose High Gain Attribute

Attribute	Gain
Gender	0.322
Car	0.171
Income level	0.171

Gender

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Travel Cost



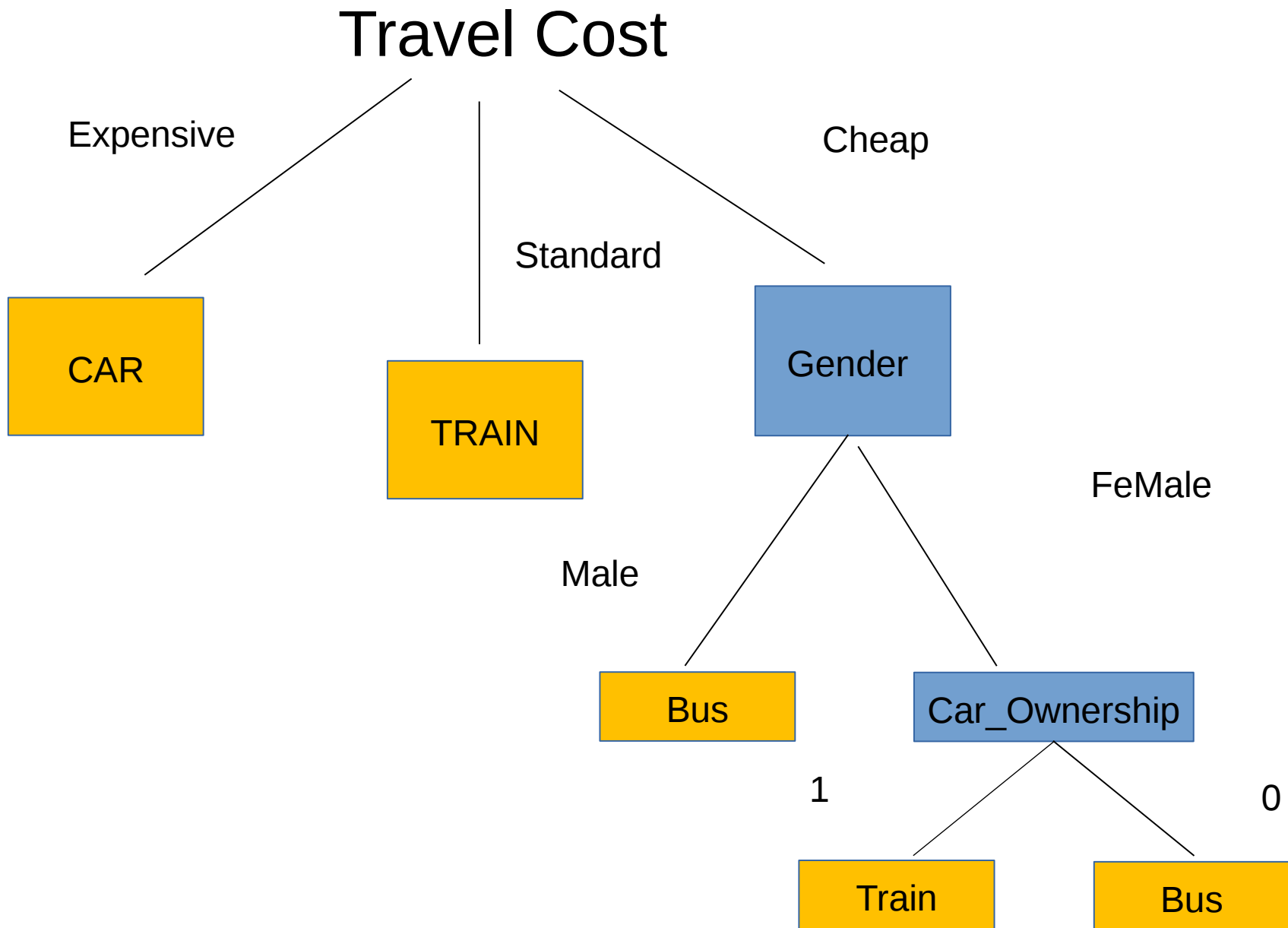
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M	2	EXPENSIVE	MEDIUM	CAR
F	2	EXPENSIVE	HIGH	CAR

$$\text{Info}(D) = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1$$

$$\text{Info Gain Car_Ownership} = 1$$

$$\text{Info Gain Income level} = 1$$

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Classify the following samples

Gender	Car Ownership	Travel cost	Income Level	Transportation
M	1	STANDARD	HIGH	TRAIN
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F	1	CHEAP	HIGH	TRAIN